

IMPACT OF DYNAMIC LINE RATING ON THE GERMAN POWER SYSTEM

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ABSTRACT

The deployment of new renewable energy sources far from load centers is leading to an increasing need for grid expansion. A notable example can be found in Germany, where most of the wind energy is generated in the north and afterwards transmitted to the industrial south [1]. Whereas the current federal government plans to add 130 GW wind power capacity until 2030, only a few grid expansion projects are in the pipeline [2]. Moreover, most of these projects have been delayed due to administrative problems and protest activities against grid expansion[3]. This trend will likely exacerbate transmission system congestion, which may lead to more grid instability and curtailment.

The transmission capacity of an overhead transmission line directly depends on its ampacity. Traditionally, transmission line ampacity is calculated at unfavorable, static weather conditions [4], referred to as Static Line Rating (SLR). By design, SLR underestimates the ampacity of a transmission line and, when implemented in practice, leads to an overestimated transmission capacity. In contrast, Dynamic Line Rating (DLR) calculates the line ampacity taking into account the prevailing weather conditions. This results in key benefits in cost-efficiency, congestion reduction and better wind power integration [5]. So far, the literature provides case studies considering DLR in simulations [4] or in small scale Energy System Optimization Models (ESOMs) [6–8]. However, it is lacking of an extensive evaluation and assessment of DLR being applied to a higher scale ESOM e.g. national scale.

This paper fills this gap. We model and analyze the impact of DLR on the optimal dispatch and deployment of power generators in the German electricity sector. The underlying model is created from the PyPSA-Eur framework [9] consisting of 40 representative nodes and considering a time horizon of one year with an hourly resolution. The DLR uses the IEEE standard [10] with highly resolved historical weather data implemented in the Atlite software [11]. The advantages of the DLR are illustrated by comparison with an SLR-based system. Therefore we analyze and compare curtailment, grid congestion as well as the optimal generator deployment. Preliminary results show a cost reduction of 5-10 % when applying DLR instead of SLR while using the same transmission system. Furthermore, DLR leads to approx. 15 % higher wind power deployment and approx. 75 % higher average power flow compared to SLR.

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